

A cubed-radical sum below 247/2017

two proofs: telescoping after AM–GM, and Jensen against $x^{-3/2}$

$$\boxed{\frac{1}{(1 + \sqrt{3})^3} + \frac{1}{(\sqrt{3} + \sqrt{5})^3} + \cdots + \frac{1}{(\sqrt{2013} + \sqrt{2015})^3} < \frac{247}{2017}}.$$

Overview. Write

$$x_k = \sqrt{2k-1}, \quad k = 1, \dots, 1009,$$

so the sum is $\sum_{k=1}^{1007} (x_k + x_{k+1})^{-3}$ with $x_1 = 1$ and $x_{1008} = \sqrt{2015}$. The whole problem rests on the identity

$$x_{k+1}^2 - x_k^2 = 2 \quad \Longleftrightarrow \quad (x_{k+1} - x_k)(x_{k+1} + x_k) = 2, \quad (1)$$

which converts a sum of *reciprocals* into a sum of *differences*, the raw material of a telescope. Concretely

$$\frac{1}{x_k + x_{k+1}} = \frac{x_{k+1} - x_k}{2}, \quad \text{hence} \quad \frac{1}{(x_k + x_{k+1})^3} = \frac{(x_{k+1} - x_k)^3}{8}. \quad (2)$$

From here two routes lead to the bound, and both end at the same place:

(A) AM–GM on $(x_k + x_{k+1})^2 > 4x_k x_{k+1}$ turns each term into $\frac{1}{8} \left(\frac{1}{x_k} - \frac{1}{x_{k+1}} \right)$, which telescopes exactly.

(B) Jensen for $t \mapsto t^3$ compares $(x_{k+1} - x_k)^3$ with $\int_{2k-1}^{2k+1} x^{-3/2} dx$, which telescopes as an integral.

The target constant is engineered: $\frac{247}{2017} = \frac{1}{8} \left(1 - \frac{41}{2017} \right)$, and $\frac{41}{2017}$ is a deliberately weakened form of $\frac{1}{\sqrt{2017}}$, since $41 = \lfloor \sqrt{2017} \rfloor$.

1. Notation and the basic identity

Set $x_k = \sqrt{2k-1}$. Then $x_1 = 1$, $x_2 = \sqrt{3}$, and the sum in question is

$$\Sigma = \sum_{k=1}^{1007} \frac{1}{(x_k + x_{k+1})^3}, \quad x_k + x_{k+1} = \sqrt{2k-1} + \sqrt{2k+1},$$

the last term being $k = 1007$, i.e. $(\sqrt{2013} + \sqrt{2015})^{-3}$, as stated.

Since $x_{k+1}^2 - x_k^2 = (2k+1) - (2k-1) = 2$, we obtain (1) and hence (2). This already rationalises the denominators: instead of a sum of reciprocals of sums of surds, we have a sum of cubes of small differences.

2. Proof A: AM–GM and an exact telescope

For $k \geq 1$ the numbers $x_k \neq x_{k+1}$ are positive, so AM–GM is strict:

$$\frac{x_k + x_{k+1}}{2} > \sqrt{x_k x_{k+1}} \quad \Longrightarrow \quad (x_k + x_{k+1})^2 > 4x_k x_{k+1} \quad \Longrightarrow \quad \frac{1}{(x_k + x_{k+1})^2} < \frac{1}{4x_k x_{k+1}}. \quad (3)$$

Multiply (3) by $\frac{1}{x_k + x_{k+1}}$ and use (2):

$$\frac{1}{(x_k + x_{k+1})^3} < \frac{1}{4x_k x_{k+1} (x_k + x_{k+1})} = \frac{x_{k+1} - x_k}{8x_k x_{k+1}} = \frac{1}{8} \left(\frac{1}{x_k} - \frac{1}{x_{k+1}} \right). \quad (4)$$

The right-hand side is a pure difference: the sum telescopes.

It is convenient to add one further term, which is positive and therefore harmless, so that the telescope ends at $x_{1009} = \sqrt{2017}$, the number appearing in the answer:

$$\Sigma < \sum_{k=1}^{1008} \frac{1}{(x_k + x_{k+1})^3} < \frac{1}{8} \sum_{k=1}^{1008} \left(\frac{1}{x_k} - \frac{1}{x_{k+1}} \right) = \frac{1}{8} \left(\frac{1}{x_1} - \frac{1}{x_{1009}} \right) = \frac{1}{8} \left(1 - \frac{1}{\sqrt{2017}} \right). \quad (5)$$

It remains to weaken $\frac{1}{\sqrt{2017}}$ into a rational number. Since $41^2 = 1681 < 2017$, we have $\sqrt{2017} > 41$ and therefore

$$\frac{1}{\sqrt{2017}} = \frac{\sqrt{2017}}{2017} > \frac{41}{2017}.$$

Subtracting a larger quantity gives a smaller result, so from (5)

$$\Sigma < \frac{1}{8} \left(1 - \frac{41}{2017} \right) = \frac{1}{8} \cdot \frac{1976}{2017} = \frac{247}{2017},$$

using $1976 = 8 \cdot 247$. $\square$

3. Proof B: Jensen against the integral of $x^{-3/2}$

The second proof replaces AM–GM by convexity and the telescope by the additivity of an integral.

Since $\frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}}$,

$$x_{k+1} - x_k = \int_{2k-1}^{2k+1} \frac{dx}{2\sqrt{x}}, \quad (6)$$

an integral over an interval of length 2. Let $g(x) = \frac{1}{2\sqrt{x}}$ and apply Jensen's inequality in integral form to the convex function $\Phi(t) = t^3$ on $(0, \infty)$:

$$\Phi \left(\frac{1}{b-a} \int_a^b g \right) \leq \frac{1}{b-a} \int_a^b \Phi(g),$$

with strict inequality because g is non-constant and Φ is strictly convex. With $a = 2k - 1$, $b = 2k + 1$, $b - a = 2$, and using (6),

$$\left(\frac{x_{k+1} - x_k}{2} \right)^3 < \frac{1}{2} \int_{2k-1}^{2k+1} \frac{dx}{8x^{3/2}} = \frac{1}{16} \int_{2k-1}^{2k+1} x^{-3/2} dx.$$

By (2) the left-hand side is exactly $\frac{1}{(x_k + x_{k+1})^3}$, so

$$\boxed{\frac{1}{(x_k + x_{k+1})^3} < \frac{1}{16} \int_{2k-1}^{2k+1} x^{-3/2} dx.} \quad (7)$$

Summing (7) for $k = 1, \dots, 1007$, the intervals $[1, 3], [3, 5], \dots, [2013, 2015]$ tile $[1, 2015]$ exactly:

$$\Sigma < \frac{1}{16} \int_1^{2015} x^{-3/2} dx = \frac{1}{16} \left[-2x^{-1/2} \right]_1^{2015} = \frac{1}{8} \left(1 - \frac{1}{\sqrt{2015}} \right). \quad (8)$$

Finally $45^2 = 2025 > 2015$ gives $\sqrt{2015} < 45$, so

$$\frac{1}{\sqrt{2015}} > \frac{1}{45} = 0.0222\dots > \frac{41}{2017} = 0.0203\dots,$$

and (8) yields

$$\Sigma < \frac{1}{8} \left(1 - \frac{41}{2017} \right) = \frac{247}{2017}. \quad \square$$

4. How much room there is

Both proofs are comfortable. The table below shows by how much.

Quantity	Exact form	Value
The sum Σ itself	–	0.1180514...
Proof B bound, (8)	$\frac{1}{8}\left(1 - \frac{1}{\sqrt{2015}}\right)$	0.1222153...
Proof A bound, (5)	$\frac{1}{8}\left(1 - \frac{1}{\sqrt{2017}}\right)$	0.1222167...
Target	$\frac{247}{2017}$	0.1224591...

So the analytic bounds give about 0.12222, against a target of 0.12246: the final rational weakening costs only $2.4 \cdot 10^{-4}$, while the inequality itself has about 3.5% of slack. The margin is thin enough that the crude step, replacing $\sqrt{2017}$ by 41, must be done in the right direction, but comfortable enough that no sharper estimate is needed.

Note that the two bounds (5) and (8) are numerically almost identical even though they were obtained by different means and end at different square roots. That is not a coincidence: both are the same telescoping estimate of $\sum \frac{1}{8}(x_k^{-1} - x_{k+1}^{-1})$, one performed discretely and one performed as $\frac{1}{16} \int x^{-3/2}$, and the two differ only by one endpoint's worth of the tail.

5. What each proof really uses

	Proof A	Proof B
Key inequality	AM–GM: $(x_k + x_{k+1})^2 > 4x_k x_{k+1}$	Jensen: $(\text{mean of } g)^3 < \text{mean of } g^3$
Telescoping device	$\frac{x_{k+1} - x_k}{x_k x_{k+1}} = \frac{1}{x_k} - \frac{1}{x_{k+1}}$	additivity of $\int_1^{2015}$ over the intervals $[2k-1, 2k+1]$
Where the exponent 3 enters	once, as the extra factor $\frac{1}{x_k + x_{k+1}}$ beyond the squared AM–GM	directly, as the convex $\Phi(t) = t^3$
Generalises to exponent p ?	only for $p = 3$ in this clean form	yes: $\frac{1}{(x_k + x_{k+1})^p} < \frac{1}{2^{p+1}} \int_{2k-1}^{2k+1} x^{-p/2} dx$

The last row is the real advantage of Proof B. Repeating Section 3 verbatim with $\Phi(t) = t^p$, $p > 1$, gives for every n

$$\sum_{k=1}^n \frac{1}{(\sqrt{2k-1} + \sqrt{2k+1})^p} < \frac{1}{2^{p+1}} \int_1^{2n+1} x^{-p/2} dx = \frac{1}{2^{p+1}} \cdot \frac{2}{p-2} (1 - (2n+1)^{1-p/2})$$

for $p \neq 2$, so the series converges precisely for $p > 2$; at $p = 3$ the bound is $\frac{1}{8}(1 - (2n+1)^{-1/2})$, recovering (8). The exponent 3 in the problem is thus the smallest odd integer for which the sum converges, and the closed form $\frac{1}{8}$ of the limiting bound is what makes a clean rational target possible.